KL UNIVERSITY · ENGINEERING CAPSTONE PROJECT SPECIFICATION

DSCI-41 — Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video

Canonical ID
DSCI-41

Dominant specialization
AI Systems for Visual Intelligence

Technical domain
Visual Intelligence Systems

Strongest impact
Industry Impact

Reference source: Zindi · Zindi
https://zindi.africa/competitions/barbados-traffic-analysis-challenge

READ THIS BEFORE CHOOSING THE PROJECT

Two-semester capstone project charter

CP1 + CP2 · complete engineering system

PROBLEM TO SOLVE

Many cities have roadside cameras but no traffic counts, because turning video into reliable vehicle counts means handling oblique camera angles, occlusion in dense traffic, mixed vehicle types and weather that degrades the image. The consequence is an unreliable visual result that can fail silently under realistic scene and sensor conditions.

WHO BENEFITS AND WHAT OWNERSHIP IS ESTABLISHED

Primary beneficiaries are system operators and reliability engineers; the cited source establishes context or source custody but does not identify an operational problem owner. No sponsor, funder or supporting organisation for the KL student implementation is stated. The KL deployment stakeholder and operational decision owner must be confirmed before approval.

FINAL ACHIEVEMENT REQUIRED

Deliver roadside video traffic counting and congestion forecasting validated under occlusion, rain and night conditions.

REFERENCE TO REPRODUCE IN CP1

Handle mixed vehicle types and oblique perspective. This is the O2 reference against which every Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video contribution must be compared.

DIFFERENTIATING CONTRIBUTION IN CP2

Sustain accuracy under occlusion, rain and low light. This is the O3 technical contribution that must be implemented, isolated and evaluated separately.

COMPLETE FINAL DEMONSTRATION

Forecast near-term congestion from estimated volumes. The final demonstration must show the complete Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video workflow rather than an isolated component.

INDEPENDENT ACCEPTANCE

Validate counting and forecasting against reference counts across conditions. The result must be independently checked under representative, boundary and failure conditions.

WHY THIS IS CAPSTONE-WORTHY

This is a two-semester culminating project because the team must establish the O2 reference “Handle mixed vehicle types and oblique perspective”, design and evaluate the O3 contribution “Accuracy under occlusion, rain and low light”, integrate the O4 system “Forecast near-term congestion from estimated volumes”, and defend the O5 result under independent, boundary and failure conditions.

Problem evidence and approval condition

DIRECT CHALLENGE OR BENCHMARK BRIEF

Evidence locator: Roadside video traffic analytics · **Open cited source**

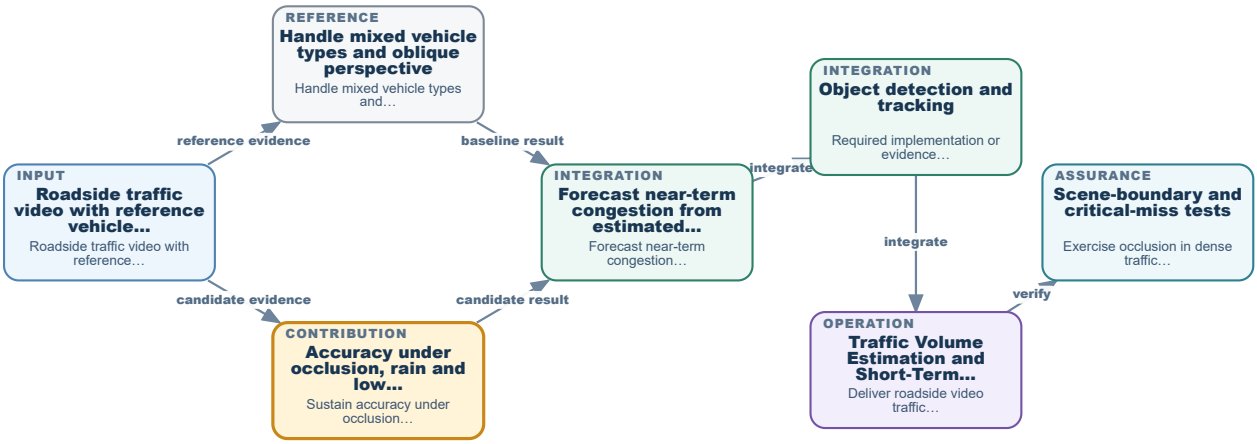
What the source establishes: The cited source establishes a reported problem, challenge, benchmark or technical requirement that can be reproduced and bounded for this project.

What the team must verify: Confirm the cited status and affected version, reproduce the reported behaviour or benchmark, and record any difference between the source and the approved project scope.

Pictorial project specification

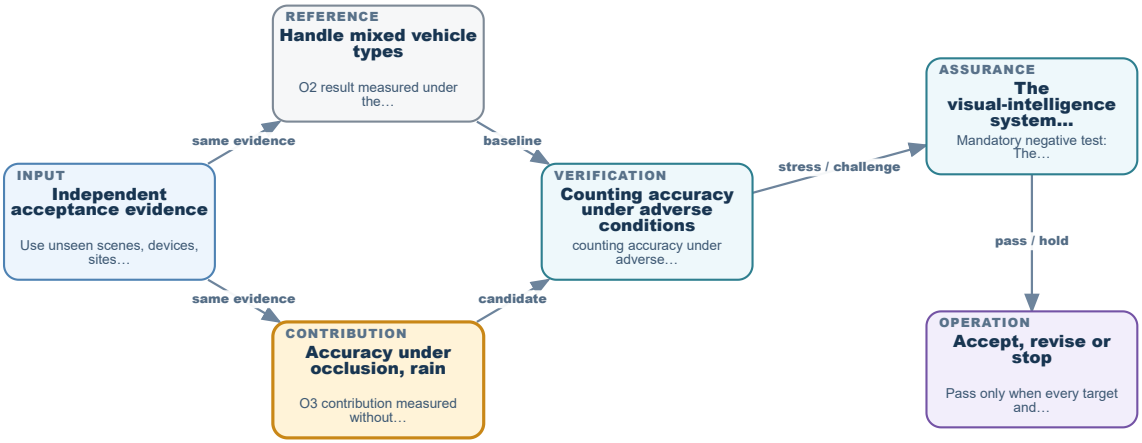
comparative vision

A. System / physical architecture — what the team must build and connect



Every arrow is an interface or evidence transfer that must be implemented and tested.

B. Validation architecture — how the team must prove the result



Every arrow is an interface or evidence transfer that must be implemented and tested.

Two-semester execution roadmap

Semester VII: problem + reference · Semester VIII: contribution + system + acceptance

SEMESTER VII / CAPSTONE PROJECT I

O1

Validate the problem and freeze the engineering contract

Build vehicle detection and counting from roadside video under a fixed camera geometry. Verify the beneficiary, source claim, system boundary, evidence access, interfaces, hazards and acceptance rules before implementation proceeds.

CP1 Gate 1 — the problem, scope, evidence and acceptance plan are approved.

Mandatory outcome and evidence

A baseline vehicle counter. Required evidence: source-to-claim map, stakeholder or reproduction evidence, requirements, architecture decision record, risk register and capture and annotation checks, geometry consistency, scene slices and confidence evidence.

O2

Build the reproducible reference and vertical slice

Handle mixed vehicle types and oblique perspective. Compare at least two feasible reference approaches, select one through a documented trade-off and measure it under AC-1 and AC-4.

CP1 Gate 2 — a reproducible reference and an integrated vertical slice are demonstrated.

Mandatory outcome and evidence

Perspective- and class-robust counting. Deliver executable fixtures, an independent oracle, unit tests, resource profile, failure log and baseline values for counting accuracy under adverse conditions; congestion forecast error; cross-camera generalisation; critical-scene failure rate (%).

SEMESTER VIII / CAPSTONE PROJECT II

O3

Engineer and prove the differentiating contribution

Sustain accuracy under occlusion, rain and low light. Compare it with O2 using the same evidence and resource envelope; include ablation, sensitivity, trade-off or controlled experiments appropriate to the project.

CP2 Gate 3 — the contribution is technically defensible and its trade-offs are measured.

Mandatory outcome and evidence

Counting accuracy under adverse conditions. Report baseline-versus-candidate results, uncertainty, condition-wise failures and a defensible positive or negative conclusion. Retain the selected configuration, experimental controls, raw evidence and a trace from each claim to its acceptance result.

04

Integrate the complete operable system

Forecast near-term congestion from estimated volumes. Connect the evidence path, reference and contribution, user or operator interface, configuration, security or safety controls, telemetry, degraded mode and recovery.

CP2 Gate 4 — the complete system operates normally, degrades safely and recovers.

Mandatory outcome and evidence

Short-term congestion forecasts. Every component and interface must have unit, integration and system evidence in a representative environment. Retain configuration, interface contracts, telemetry, degraded-mode behaviour and recovery proof.

05

Complete independent acceptance and impact validation

Validate counting and forecasting against reference counts across conditions. Execute AC-1 to AC-4 and all 5 negative tests; obtain independent domain, stakeholder or technical review.

CP2 Final Gate — every target and guardrail passes, or the project is explicitly held or revised.

Mandatory outcome and evidence

A validation report across conditions. Publish baseline, target and achieved values, limitations, residual risks, reproducibility package, operating guide, final report, demonstration video and individual contribution evidence.

Functional requirements

- FR-1 — Ingest or acquire “Roadside traffic video with reference vehicle counts”; validate version, provenance, schema or protocol, units and project-specific quality gates before use.
- FR-2 — Reproduce “Handle mixed vehicle types and oblique perspective” through a one-command, one-protocol or one-test-run reference path with an independent expected-result oracle.
- FR-3 — Implement “Accuracy under occlusion, rain and low light” as a separately testable subsystem whose configuration and evidence are versioned.
- FR-4 — Implement and verify “Forecast near-term congestion from estimated volumes” and its input, output, error, configuration and telemetry contracts.
- FR-5 — Implement and verify “Object detection and tracking” and its input, output, error, configuration and telemetry contracts.
- FR-6 — Deliver “Traffic Volume Estimation and Short-Term Congestion Forecasting” as the complete demonstrable result; no isolated notebook, model, function or dashboard counts as completion.
- FR-7 — Detect the condition described by NT-1, invoke the documented safe or degraded response and retain recovery evidence.

Non-functional and domain-assurance requirements

- NFR-1 — Performance: meet every accepted KPI target under AC-1 and AC-4, including tail values and resource cost where applicable.
- NFR-2 — Reliability and recovery: pass every mandatory negative test; a critical failure cannot be hidden by an average.
- NFR-3 — Domain assurance: Capture, geometry, annotation and transformation consistency shall be machine-checkable.
- NFR-4 — Evidence quality: Small, rare and safety-critical objects or scenes shall be reported separately from aggregate accuracy.
- NFR-5 — Maintainability: use version control, automated tests, documented interfaces, configuration validation and a clean-environment build or protocol.
- NFR-6 — Operability: provide logs, measurements, health or quality indicators, operator guidance, degraded mode and a verified recovery procedure.

System boundary and permitted scope

INCLUDED

- Reproducible reference: Handle mixed vehicle types and oblique perspective. This is the O2 reference against which every Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video contribution must be compared
- Differentiating contribution: Sustain accuracy under occlusion, rain and low light. This is the O3 technical contribution that must be implemented, isolated and evaluated separately
- Integrated result: Forecast near-term congestion from estimated volumes. The final demonstration must show the complete Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video workflow rather than an isolated component
- Independent acceptance: Validate counting and forecasting against reference counts across conditions. The result must be independently checked under representative, boundary and failure conditions

EXPLICITLY EXCLUDED

- out of scope.
- A notebook, isolated model, single-function prototype, dashboard-only demonstration or unvalidated library integration.
- Production certification, universal generalisation and unattended high-stakes operation outside the validated envelope.
- An isolated notebook, model, function, dashboard or library integration without the complete operating and validation workflow.

DEPENDENCY ASSUMPTION

Access to every required dataset, sample, device, tool, reviewer, licence and approval must be confirmed at the project-approval review; unavailable dependencies require an approved scope revision.

Team, prerequisites, tools and feasibility

RECOMMENDED TEAM

A team of 3–4 students must assign named responsibility for image or scene evidence leadership, vision model and geometry leadership, application and deployment leadership, critical-case validation leadership in Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video; each member must own reviewable implementation and evidence.

PREREQUISITES

Computer vision, tracking, forecasting.

PRINCIPAL TOOLS

- object detection and tracking
- perspective handling
- short-term forecasting
- Python

FACILITY / COMPUTE

GPU and storage for the declared image/video resolution; use tiling, sampling or a smaller model if the full corpus is infeasible. Principal access must include object detection and tracking, perspective handling, short-term forecasting, Python.

FALLBACK PATH

Use a representative, leakage-resistant subset or tiled workflow while preserving scene diversity and the independent validation split. It may reduce scale but must retain O2, O3, O4 and O5.

Acceptance conditions

AC-1 **Representative operation**

Validate counting and forecasting against reference counts across conditions.

AC-2 **Boundary and failure operation**

Exercise occlusion in dense traffic; night and rain degradation; camera-specific calibration.

AC-3 **Independent acceptance evidence**

Use unseen scenes, devices, sites or conditions and report small, rare and safety-critical cases separately.

AC-4 **Frozen resource envelope**

Use the same approved versions, workload, hardware or facility limits and resource budget for reference and final comparisons; principal tools are object detection and tracking, perspective handling, short-term forecasting, Python.

KPI baseline, target, method and pass contract

ID	MEASURE	BASELINE	DIRECTION	UNIT	TARGET / PASS RULE	CONDITIONS	MEASUREMENT METHOD	GUARDRAIL
KPI-1	**counting accuracy under adverse conditions**	O2 reference measured with the same evidence, workload and resource envelope	higher is better	%	≥ O2 reference + 3 percentage points, or within 1 point while improving the primary resource KPI by ≥ 20%	AC-1, AC-3	Use the leakage-resistant independent partition; report condition-wise values, 95% confidence intervals and the raw confusion or error evidence.	No critical subgroup, scene or operating condition may be hidden by an aggregate result.
KPI-2	**congestion forecast error**	O2 reference measured with the same evidence, workload and resource envelope	lower is better	dimensionless score on a 0–100 rubric frozen before O2	≥ 80/100 and at least 5 points higher than the O2 reference	AC-1, AC-3	Before O2, freeze a 0–100 rubric with at least five observable criteria and named 0, 50, 80 and 100 anchors; use two independent raters and report agreement plus condition-wise scores.	No critical subgroup, scene or operating condition may be hidden by an aggregate result.
KPI-3	**cross-camera generalisation**	O2 reference measured with the same evidence, workload and resource envelope	higher is better	dimensionless score on a 0–100 rubric frozen before O2	≥ 80/100 and at least 5 points higher than the O2 reference	AC-1, AC-3	Before O2, freeze a 0–100 rubric with at least five observable criteria and named 0, 50, 80 and 100 anchors; use two independent raters and report agreement plus condition-wise scores.	No critical subgroup, scene or operating condition may be hidden by an aggregate result.
KPI-4	**critical-scene failure rate (%)**	O2 reference measured with the same evidence, workload and resource envelope	lower is better	%	≤ 0.80 × O2 reference; no critical condition may be worse than the reference	AC-1, AC-3	Use the leakage-resistant independent partition; report condition-wise values, 95% confidence intervals and the raw confusion or error evidence.	No critical subgroup, scene or operating condition may be hidden by an aggregate result.

ID	MEASURE	BASELINE	DIRECTION	UNIT	TARGET / PASS RULE	CONDITIONS	MEASUREMENT METHOD	GUARDRAIL
KPI-5	**cross-condition performance gap (%)**	O2 reference measured with the same evidence, workload and resource envelope	lower is better	%	≤ 0.80 × O2 reference; no critical condition may be worse than the reference	AC-1, AC-3	Use the leakage-resistant independent partition; report condition-wise values, 95% confidence intervals and the raw confusion or error evidence.	No critical subgroup, scene or operating condition may be hidden by an aggregate result.
KPI-6	**inference latency p95 (ms)**	O2 reference measured with the same evidence, workload and resource envelope	lower is better	ms	p95 ≤ 0.80 × O2 reference at equal or better functional quality	AC-1, AC-4	Warm the system, run at least 30 measured trials per condition with a monotonic clock, and report p50, p95, p99 and failures.	The primary correctness, safety or quality metric may not regress by more than the approved non-inferiority margin.

Mandatory negative-test and recovery campaign

NT-1 The visual-intelligence system encounters occlusion in dense traffic during the Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video operating or validation workflow.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The visual-intelligence system encounters occlusion in dense traffic during the Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video operating or validation workflow”.

Expected safe behaviour: The system must detect the condition, abstain or enter the approved degraded mode, label the affected result and prevent silent use outside the validated envelope.

Recovery: Restore the accepted preprocessing and model bundle; rerun the affected scene class and retain the failure example.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT-2 The visual-intelligence system encounters night and rain degradation during the Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video operating or validation workflow.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The visual-intelligence system encounters night and rain degradation during the Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video operating or validation workflow”.

Expected safe behaviour: The system must detect the condition, abstain or enter the approved degraded mode, label the affected result and prevent silent use outside the validated envelope.

Recovery: Restore the accepted preprocessing and model bundle; rerun the affected scene class and retain the failure example.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT-3 The visual-intelligence system encounters camera-specific calibration during the Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video operating or validation workflow.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The visual-intelligence system encounters camera-specific calibration during the Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video operating or validation workflow”.

Expected safe behaviour: The system must detect the condition, abstain or enter the approved degraded mode, label the affected result and prevent silent use outside the validated envelope.

Recovery: Restore the accepted preprocessing and model bundle; rerun the affected scene class and retain the failure example.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT-4 The visual-intelligence system encounters average accuracy hides failure on small, rare or safety-critical targets in Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video during the Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video operating or validation workflow.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The visual-intelligence system encounters average accuracy hides failure on small, rare or safety-critical targets in Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video during the Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video operating or validation workflow”.

Expected safe behaviour: The system must detect the condition, abstain or enter the approved degraded mode, label the affected result and prevent silent use outside the validated envelope.

Recovery: Restore the accepted preprocessing and model bundle; rerun the affected scene class and retain the failure example.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT-5 Pre-processing or augmentation leaks information across evidence partitions in Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “Pre-processing or augmentation leaks information across evidence partitions in Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video”.

Expected safe behaviour: The system must detect the condition, abstain or enter the approved degraded mode, label the affected result and prevent silent use outside the validated envelope.

Recovery: Restore the accepted preprocessing and model bundle; rerun the affected scene class and retain the failure example.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

Evidence and reproducibility contract

PRIMARY EVIDENCE

Roadside traffic video with reference vehicle counts across weather and lighting

PROTOCOL AND QUALITY

Version the cited evidence and project inputs, reproduce “Handle mixed vehicle types and oblique perspective”, implement “Sustain accuracy under occlusion, rain and low light”, and compare both under AC-1 to AC-4 without changing the approved evidence or resource envelope. Check annotation quality, scene/device/site separation, geometry or registration, rare targets and critical-miss coverage.

PROVENANCE AND INDEPENDENT PARTITION

The Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video evidence manifest must record source URL, access date, licence or permission, owner/custodian, record/sample/fixture identifiers, exclusions, transformations, configuration and the objective or KPI using each item. Reserve the AC-3 evidence before O3 tuning and separate it by the independent unit appropriate to visual-intelligence engineering; no transformation may leak acceptance information into development.

REPRODUCIBILITY

A clean environment must rebuild the O2 reference, O3 contribution, integrated system and acceptance evidence for Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video using the version-pinned toolchain (object detection and tracking, perspective handling, short-term forecasting, Python), seeds/configuration and an automated or fully scripted runbook.

Mandatory final deliverables

- D1 — Approved Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video problem charter, requirements, system boundary, architecture and source-to-claim map.
- D2 — Reproducible Handle mixed vehicle types and oblique perspective with fixtures or protocol, oracle, tests, baseline results and resource profile.
- D3 — Independently testable Accuracy under occlusion, rain and low light with design rationale, comparative evidence, trade-offs and limitations.
- D4 — Complete Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video prototype or experimental system with configuration, interfaces, telemetry, degraded operation and recovery.
- D5 — Automated or documented acceptance harness covering AC-1 to AC-4, all KPIs and every negative test.
- D6 — Versioned repository or protocol package, evidence manifest, installation/operating guide, API/interface documentation and reproducibility instructions.
- D7 — Final report, research paper or technical article, poster, portfolio case study, demonstration video, video logs and individual contribution/viva evidence.

PROJECT-SPECIFIC REFERENCE RECORD

Reference metadata and classification

This section explains how this project is sourced, classified, staffed, governed and measured.
Every statement must remain consistent with the approved project charter and evidence.

STUDENT REFERENCE RECORD

1 Source, problem issuers and custodianship and evidence boundary

EVIDENCE STATUS

Direct challenge or benchmark record; named issuer retained where stated

Direct challenge or benchmark brief

SOURCE PLATFORM / PUBLISHER

Zindi

Zindi

EXACT SOURCE RECORD TYPE

Challenge, competition or benchmark brief

VERIFIED SOURCE AUTHORITY

Zindi

PROGRAMME STEWARD

Zindi

PROBLEM ISSUER / UPSTREAM CUSTODIAN

Not identified by the cited record

OPERATIONAL PROBLEM-OWNER STATUS

The cited record is direct, but it does not name an organisation that can safely be presented as the operational problem owner.

EXPLICIT SPONSORSHIP / FUNDING / SUPPORT STATUS

No separate sponsor, funder or supporting organisation is explicitly identified in the retained cited record.

EVIDENCE LOCATOR

Roadside video traffic analytics

ATTRIBUTION BASIS

The cited URL is hosted at zindi.africa; the retained source label is “Zindi”. Platform, issuer, custodian and sponsor roles are recorded separately.

WHAT THE SOURCE ESTABLISHES

The cited source establishes a reported problem, challenge, benchmark or technical requirement that can be reproduced and bounded for this project.

WHAT MUST BE RECHECKED BEFORE APPROVAL

Confirm the cited status and affected version, reproduce the reported behaviour or benchmark, and record any difference between the source and the approved project scope.

CLAIM BOUNDARY

The source supports the stated starting problem or brief. It does not establish that the issue remains open, that every proposed extension is requested, that a named organisation sponsors the student team, or that the final solution is production-ready.

RETAINED ORIGINAL SOURCE LABEL

Zindi

CASE-AUDITED ATTRIBUTED ORGANISATIONS

Zindi

Source authority · Commercial company or industry body

Technology, Software & Cloud
Programme steward or publisher for the cited Zindi record; the platform itself is not treated as the operational problem owner.

2 Classification, specialization rationale and AI applicability

ENGINEERING FAMILY

visual-intelligence engineering

TECHNICAL DOMAIN

Visual Intelligence Systems

APPLICATION SECTOR

Automotive, Mobility & Transportation

ISSUER / CUSTODIAN SECTOR

Not applicable — no direct issuer or custodian is claimed

ATTRIBUTED-ORGANISATION CLASSIFICATION

Commercial company or industry body

CANONICAL PROJECT ID

DSCI-41

Dominant specialization · AI Systems for Visual Intelligence

For Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video, this is the dominant specialization because the principal assessed work is the O3 contribution: “Sustain accuracy under occlusion, rain and low light”. Students must compare it with the O2 reference: “Handle mixed vehicle types and oblique perspective”, then integrate the O4 operating workflow: “Forecast near-term congestion from estimated volumes”. The resulting artefact is assessed as an end-to-end Visual Intelligence Systems capstone. Curriculum anchors: Visual recognition and scene understanding; AI driven video processing.

Related specialization · Data Science & Big Data Analytics

For Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video, Data Science & Big Data Analytics is a related specialization because its methods, interfaces or assurance concerns support the O4 workflow: “Forecast near-term congestion from estimated volumes”. It is supporting scope; the principal assessed contribution remains “Sustain accuracy under occlusion, rain and low light” under AI Systems for Visual Intelligence. Curriculum anchors: Digital video processing.

Not required for the core capstone

The approved O3 contribution—“Sustain accuracy under occlusion, rain and low light”—is primarily Visual Intelligence Systems work and can be completed without an AI/ML subsystem.

Boundary: AI/ML/GenAI may be added only through an approved comparative hypothesis; it may not replace the required engineering mechanism, calculations, tests or domain evidence

3 Team, prerequisites, facilities and feasibility

RECOMMENDED TEAM SIZE

3–4 students

PREREQUISITE COMPETENCE

Computer vision, tracking, forecasting.

FACILITY AND COMPUTE

GPU and storage for the declared image/video resolution; use tiling, sampling or a smaller model if the full corpus is infeasible. Principal access must include object detection and tracking, perspective handling, short-term forecasting, Python.

ACCESS DECISION

Before approving Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video, confirm the cited problem status and reproduce the source-supported starting condition; record access to object detection and tracking, perspective handling, short-term forecasting, Python, all data or physical evidence, required permissions and an independent reviewer for counting accuracy under adverse conditions and congestion forecast error.

FEASIBLE FALLBACK

Use a representative, leakage-resistant subset or tiled workflow while preserving scene diversity and the independent validation split. It may reduce scale but must retain O2, O3, O4 and O5.

NAMED ROLE OWNERSHIP AND COMPULSORY EVIDENCE

Image or scene evidence lead

Own the source-to-claim map, access confirmation and evidence needed to execute the O2 reference: “Handle mixed vehicle type and oblique per pective. Thi i the O2 reference again…”.

Evidence manifest and O2 reproduction

Vision model and geometry lead

Own the design, implementation and isolated evaluation of the O3 contribution: “Su tain accuracy under occlu ion, rain and low light. Thi i the O3…”.

O3 design record, implementation and equal-condition comparison

Application and deployment lead

Own all interfaces, telemetry and recovery needed for the O4 operating workflow: “Foreca t near-term conge tion from e timated volume . The final demon tration…”.

O4 integration tests, operating guide and recovery evidence

Critical-case validation lead

Own condition-wise counting accuracy under adverse conditions and congestion forecast error results, every negative test and intended-versus-actual impact.

O5 acceptance dossier, independent review and individual viva

4 Controlled technologies and their required roles

CONTROLLED TECHNOLOGY	CATEGORY	REQUIRED USE IN THIS PROJECT
Time-Series / Forecasting Libraries	AI & ML	Implements and evaluates the O3 contribution—“Su tain accuracy under occlu ion, rain and low light. Thi…”—against the O2 reference under identical evidence conditions.
OpenCV / Vision Models	AI & ML	Implements and evaluates the O3 contribution—“Su tain accuracy under occlu ion, rain and low light. Thi…”—against the O2 reference under identical evidence conditions.
Python	Domain Tools	Implements the O2 reference, the O3 contribution and the automated evidence harness for this project.
SQL / PostgreSQL	Data & Analytics	Stores, transforms, versions or serves evidence for the O4 workflow—“Foreca t near-term conge tion from e timated volume ….”—with lineage and reconciliation.
Apache Spark	Data & Analytics	Stores, transforms, versions or serves evidence for the O4 workflow—“Foreca t near-term conge tion from e timated volume ….”—with lineage and reconciliation.
Git / Reproducible Environments	Domain Tools	Provides a version-pinned domain toolchain for building and reproducing this project.

5 Controlled skills and the evidence that proves them

CONTROLLED SKILL	CATEGORY	EVIDENCE THAT PROVES THE SKILL
Data Science / Statistical Analysis	Data Science & Analytics	O2 reproduction plus implementation and condition-wise evaluation of the O3 contribution: “Su tain accuracy under occlu ion, rain and low light. Thi i the O3…”.

CONTROLLED SKILL	CATEGORY	EVIDENCE THAT PROVES THE SKILL
Validation / Benchmarking / Uncertainty	Professional & Transversal	Equal-condition O2–O3 comparison and independent evidence for counting accuracy under adverse conditions (≥ O2 reference + 3 percentage points, or within 1 point while improving the primary resource KPI by ≥ 20%).
Technical Documentation / Communication	Professional & Transversal	Architecture decisions, operating guide, evidence index, final report, paper and individual viva.
Computer Vision	AI & Intelligent Systems	O2 reproduction plus implementation and condition-wise evaluation of the O3 contribution: “Su tain accuracy under occlu ion, rain and low light. Thi i the O3…”.

6 SDG target alignment and claim boundary

SDG 11.2 · Safe, affordable and accessible transport

Contribution: Supports data-driven transport planning. The project contributes through this bounded deliverable: “Deliver roadside video traffic counting and congestion forecasting validated under occlusion, rain and night conditions”. The intended beneficiaries are transport operators, mobility planners and road users.

Evidence: Report counting accuracy under adverse conditions against “≥ O2 reference + 3 percentage points, or within 1 point while improving the primary resource KPI by ≥ 20%” with the relevant stakeholder, subgroup, safety or resource guardrail.

Enabling contribution only; report measured local evidence and do not claim achievement of the UN target.

SDG 9.1 · Develop sustainable, resilient infrastructure

Contribution: Improves infrastructure decisions. The project contributes through this bounded deliverable: “Deliver roadside video traffic counting and congestion forecasting validated under occlusion, rain and night conditions”. The intended beneficiaries are transport operators, mobility planners and road users.

Evidence: Report congestion forecast error against “≥ 80/100 and at least 5 points higher than the O2 reference” with the relevant stakeholder, subgroup, safety or resource guardrail.

Enabling contribution only; report measured local evidence and do not claim achievement of the UN target.

7 Weighted impact and project-specific measurement

Strongest impact: Industry Impact. Weights express the project’s intended evidence emphasis; they are not marks and do not replace the CP1/CP2 evaluation rubrics.

Graduate Impact

18%

The team must demonstrate Data Science / Statistical Analysis, Validation / Benchmarking / Uncertainty, Technical Documentation / Communication while reproducing the O2 reference and delivering the complete Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video system.

Measurement: For Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video, assess each student on the project-specific evidence for Data Science / Statistical Analysis, Validation / Benchmarking / Uncertainty, Technical Documentation / Communication, objective-to-contribution traceability, individual commits or protocol ownership, final demonstration and viva; each critical competency must meet the approved rubric threshold.

Industry Impact

24%

The integrated artefact is intended to improve the operating or decision workflow represented by “Forecast near-term congestion from estimated volumes”.

Measurement: For Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video, compare the O2 reference and final system on counting accuracy under adverse conditions (≥ O2 reference + 3 percentage points, or within 1 point while improving the primary resource KPI by ≥ 20%) and congestion forecast error (≥ 80/100 and at least 5 points higher than the O2 reference); retain operating-condition, failure, recovery and independent reviewer evidence.

Research Impact

19%

The project makes a testable comparison between the O2 reference—“Handle mixed vehicle types and oblique perspective”—and the O3 contribution—“Sustain accuracy under occlusion, rain and low light”—including negative results and declared limitations.

Measurement: For Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video, use identical evidence and resource conditions, repeat stochastic trials where applicable, report uncertainty or effect size, perform the relevant ablation or differential test and provide a clean reproduction package.

Societal Impact

22%

The intended beneficiaries are transport operators, mobility planners and road users; public value may be claimed only to the extent measured in the approved pilot or independent review.

Measurement: For Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video, report counting accuracy under adverse conditions (≥ O2 reference + 3 percentage points, or within 1 point while improving the primary resource KPI by ≥ 20%) together with one beneficiary access, safety, inclusion or usability measure and one adverse-effect or subgroup guardrail aligned to SDG 11.2.

Environmental Impact

17%

The project has an explicit resource or environmental consequence that must be compared per accepted output or functional unit.

Measurement: For Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video, measure the applicable compute, energy, storage, network, material or waste quantity per accepted result under the same O2 and final-system workload; report “not material” only with evidence.

PROJECT-SPECIFIC IMPACT MEASUREMENT PLAN

For Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video, freeze the O2 value, final target, evidence source, workload or sampling protocol and reviewer for counting accuracy under adverse conditions and congestion forecast error; re-measure after O3 integration and at O5, report uncertainty, intended-versus-actual impact, attribution limits and every unmet condition.

8 Access, licensing, safety, ethics and disclosure controls

DATA, FACILITY AND AUTHORISATION

Confirm image/video consent and licence, de-identification, annotation rights, sensitive-scene handling and model-use limitations.

LICENSING, ATTRIBUTION AND IP

Record licences and attribution for the cited source, data, models, software, hardware designs and reused artefacts; do not redistribute restricted material in the Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video evidence package.

SAFETY, ETHICS AND HUMAN OVERSIGHT

Document foreseeable harm, misuse, privacy, security, accessibility, environmental and domain-safety risks; implement prevention, degraded mode, human oversight and escalation.

AI-ASSISTED WORK AND MODEL DISCLOSURE

Disclose any AI-assisted coding, analysis or writing. It does not count as the project contribution and must be verified by the students.

SOURCE-STATUS INTERPRETATION

Recheck source availability, licence, affected version and current status before approval; do not imply maintainer endorsement or production readiness.